



C20-EC-502

7641

BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DECE – FIFTH SEMESTER EXAMINATION

MICROCONTROLLERS AND APPLICATIONS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List any six special functional registers of 8051.
2. Define the terms (a) operation code and (b) operand.
3. List any three one byte instructions of 8051.
4. State the effect on flags of 8051 due to the following instructions
MOV A, #25H
ADD A, #A1H.
5. Write an assembly language program to load accumulator with the value 65H and complement the accumulator.
6. State key bouncing problem.
7. Write an ALP to blink an LED connected to P1.0 of 8051.
8. Compare any three features of RISC and CISC architectures.
9. List the important features of ARM7.
10. List any five applications of embedded system.

PART—B

8×5=40

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **eight** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** (a) Draw the functional block diagram of 8051 and explain about each block.

(OR)

- (b) Draw the pin diagram of 8051 microcontroller and specify the purpose of each pin.

- 12.** (a) Explain the SFRs associated with timer/counters of 8051.

(OR)

- (b) Explain the SFRs associated with interrupts of 8051.

- 13.** (a) Explain the Data Transfer Instructions of 8051 with examples.

(OR)

- (b) Explain the unconditional jump instructions of 8051.

- 14.** (a) Write an assembly language program to transfer a block of 10 numbers stored at internal RAM location 30H onwards to external RAM location 5000H onwards.

(OR)

- (b) Write an assembly language program to generate 1 ms delay using timer 0 in mode 1. Assume crystal oscillator frequency as 12 MHz.

- 15.** (a) Explain PUSH and POP instructions with an example.

(OR)

- (b) Define a subroutine and explain the sequence of program when

subroutine is called and executed.

PART—C

10×1=10

- Instructions :**
- (1) Answer the following question.
 - (2) The question carries **ten** marks.
 - (3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. A push button switch (SW) is connected to pin P1·0 and an LED to pin P2·7. Draw the interfacing diagram and write a program to check the status of SW and perform the following :

- (a) If SW is not pressed (SW = 0), make LED ON
- (b) If SW is pressed (SW = 1), make LED OFF

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